

# Junrong Yan

## Curriculum Vitae

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## Education

2017–2022 **Ph.D.**, *Department of Mathematics*, University Of California, Santa Barbara.  
Advisor: Xianzhe Dai

2013–2017 **B.S in Mathematics**, *School of Mathematical Sciences*, Nankai University.

## Employment

2024–Current **Zelevinsky Postdoctoral Fellow**, *Department of Mathematics*, Northeastern University.  
Mentor: Maxim Braverman

2022–2024 **Boya Postdoctoral Fellow**, *Beijing International Center for Mathematical Research*, Peking University.  
Mentor: Xiaobo Liu

## Awards and Grants

2018 Wilder Award at UCSB

2021 Grade Student Travel Grant to JMM

2022 Boya Postdoctoral Fellowship at PKU

2023 General Program in China Postdoctoral Science Foundation

2024 Zelevinsky Postdoc Travel Grant

## Research Interests

Analysis Global analysis on noncompact manifolds; spectral theory of elliptic, Dirac, and Schrödinger operators; heat-kernel asymptotics; Weyl laws and Tauberian theory; Witten deformation.

Mathematical Physics Mathematical aspects of quantum field theory; Feynman graph integrals and renormalization; holomorphic field theory; mirror symmetry; Landau–Ginzburg models; BCOV theory; CY/LG correspondence.

Geometry and Topology Differential and comparison geometry; index theory; Morse theory; analytic torsion and torsion forms; gluing formulas; higher torsion.

## Papers

- Published or accepted
- [1] Xianzhe Dai and Junrong Yan. *Witten deformation for noncompact manifolds with bounded geometry*, Journal of the Institute of Mathematics of Jussieu, 2023, 38 pages
  - [2] Xianzhe Dai and Junrong Yan. *Witten deformation on non-compact manifold: heat kernel expansion and local index theorem*, 2023, Math. Z., 28 pages
  - [3] Xinxing Tang and Junrong Yan. *CY/LG correspondence for Weil-Petersson metrics and  $tt$  structures*, Journal of Geometry and Physics, 2025, 34 pages

- [4] Fagui Li and Junrong Yan. *A first eigenvalue estimate for embedded hypersurfaces in positive Ricci curvature manifolds*. Differential Geometry and its Applications, 2026, 10 pages
- [5] Junrong Yan. *Witten deformation for non-Morse Functions and gluing formula for analytic torsions*, To appear in CAG, 39 pages

- Preprint [1] Junrong Yan. *A new analytic proof of gluing formula for analytic torsion forms*, arXiv:2301.02591, 2023, 40 pages
- [2] Martin Puchol and Junrong Yan. *Generalized Morse Functions, Excision and Higher Torsions*. arXiv:2407.17100, 2024, 107 pages
- [3] Xianzhe Dai and Junrong Yan. *Weyl Law for Schrödinger Operators on Noncompact Manifolds, Heat Kernel, and Karamata-Hardy-Littlewood Theorem*. arXiv:2504.15551, 2025, 28 pages
- [4] Maxim Braverman, Ahmad Reza Haj Saeedi Sadegh and Junrong Yan. *A Gluing Formula for the  $\mathbb{Z}_2$ -valued Index of Odd Symmetric Operators*,. arXiv:2505.07094, 2025, 18 pages
- [5] Minghao Wang and Junrong Yan. *Feynman Graph Integrals on Kähler Manifolds*. arXiv:2507.09170, 2025, 49 pages
- [6] Maxim Braverman, Xianzhe Dai and Junrong Yan. *The Classical Weyl Law for Schrödinger Operators on Complete Riemannian Manifolds*. arxiv:2605.07200, 2026, 35 pages
- [7] Minghao Wang and Junrong Yan. *Renormalizations in Holomorphic Field Theories on Kähler Manifolds*. arXiv:2608.00546, 2026, 82 pages
- [8] Junrong Yan. *The Weyl Law Meets Large-Scale Regularity on Ricci Shrinkers*. arXiv:2609.0820, 29 pages

## Academic Services

- Paper Referee SIGMA (Symmetry, Integrability and Geometry: Methods and Applications); Journal für die reine und angewandte Mathematik; Pacific Journal of Mathematics; Communications in Mathematical Physics; Journal of Geometric Analysis.
- Reviewer zbMATH Open; MathSciNet.

## Teaching Experience

### Instructor

- Northeastern University *MATH 7371: Morse Theory*, Fall 2025.
- MATH 2331: Linear Algebra*, Fall 2025.
- MATH 1342: Calculus 2 for Science and Engineering*, Fall 2024; Spring 2025.
- Peking University *Calculus 2*, Winter 2024.
- UC Santa Barbara *MATH 3B: Calculus with Applications*, Summer 2021.

### Teaching Assistant

- MATH 8: Transitions to Higher Mathematics*, Spring 2018; Fall 2018; Fall 2020.

*MATH 6B: Vector Calculus*, Winter 2019.

*MATH 34B: Calculus for Social and Life Sciences*, Spring 2019.

*MATH 6A: Vector Calculus*, Fall 2019; Winter 2020; Summer 2020; Winter 2021; Spring 2021.

#### Reader

*MATH 117: Methods of Analysis*, Fall 2017; Fall 2019.

*MATH 108B: Introduction to Linear Algebra*, Winter 2018.

### Undergraduate Mentoring (2021 UCSB Directed Reading Program)

Jiasheng Lin Morse theory and Floer homology  
Kai Yan & Gauge theory, TQFT and knot theory  
Xinzhe Li

### Selected Talks

- Aug. 2026 **Graph integral on Kähler manifolds**, *Workshop on Global Analysis on Manifolds*, Tianjin
- June 2026 **Graph integral on Kähler manifolds**, *University of Bonn*, Bonn
- June 2026 **Graph integral on Kähler manifolds**, *Universite Paris-Saclay*, Orsay
- Apr. 2026 **Graph integral on Kähler manifolds**, *Texas A&M University*, College Station
- Mar. 2026 **Graph integral on Kähler manifolds**, *George Mason University*, Washington DC
- Feb. 2026 **Weyl's law on complete manifolds**, *University of California, Santa Barbara*, Santa Barbara
- Feb. 2026 **Graph integral on Kähler manifolds**, *University of California, Irvine*, Irvine
- Jan. 2026 **Graph integral on Kähler manifolds**, *Boston University*, Boston
- Dec. 2025 **Graph integral on Kähler manifolds**, *Wuhan University*, Wuhan
- June. 2025 **Heat kernel expansion, Tauberian theory and Weyl law for Schrodinger operator on noncompact manifolds**, *AMS Special Session on Noncommutative Geometry and Applications*, St. Louis
- June. 2025 **Heat kernel expansion, Tauberian theory and Weyl law for Schrodinger operator on noncompact manifolds**, *Xiamen University*, Xiamen
- May. 2025 **Heat kernel expansion and Weyl's law for Schrödinger-type operators on noncompact manifolds**, *Tsinghua University*, Beijing
- April. 2025 **Heat kernel expansion and Weyl's law for Schrödinger-type operators on noncompact manifolds**, *University of Tokyo*, Tokyo
- Mar. 2025 **Witten deformation for non-Morse functions and gluing formulas**, *Go Fishing 2025–Possion Geometry Meeting*, St. Louis
- Dec. 2024 **Towards the comparison of higher torsion invariants**, *Around Local Index Theorem*, Paris
- Sept. 2024 **Higher torsions and Witten-type deformation**, *Northeastern University*, Boston
- Sept. 2024 **Witten deformation for non-Morse functions and gluing formulas**, *Harvard University*, Boston
- Aug. 2024 **Higher torsions and Witten-type deformation**, *Workshop on Global Analysis on Manifolds 2024*, Wuhan

- Feb. 2024 **Witten deformation for non-Morse functions and higher Cheeger-Müller theorem**, *Conference on Dirac operators: analysis and geometry*, Beijing
- Nov. 2023 **Witten deformation for non-Morse functions and gluing formulas**, *University of Freiburg*, Freiburg
- Oct. 2023 **Witten deformation for non-Morse functions and gluing formulas**, *Université Paris-Saclay*, Orsay
- Mar. 2023 **Witten deformation for non-Morse functions and gluing formulas**, *Chern Institute of Mathematics, Nankai University*, Tianjin
- Sept. 2022 **Calabi-Yau/Landau-Ginzburg correspondence for  $tt^*$  Structures**, *Peking University*, Beijing
- Jun. 2022 **Calabi-Yau/Landau-Ginzburg correspondence for  $tt^*$  Structures**, *The Chinese University of Hong Kong*, Hongkong
- Jan. 2022 **Analytic torsion on noncompact manifolds**, *AMS Special Session on Intersections of geometric analysis and mathematical physics*, Seattle
- Jan. 2021 **Heat kernel expansion and local index theorem on noncompact manifolds**, *Peking University*, Beijing
- Dec. 2020 **Heat kernel expansion and local index theorem on noncompact manifolds**, *Chern Institute of Mathematics, Nankai University*, Tianjin
- Nov. 2020 **Heat kernel expansion and local index theorem on noncompact manifolds**, *East China Normal University*, Shanghai
- Apr. 2020 **Witten deformation on noncompact manifolds and mirror symmetry for Landau-Ginzburg model**, *UCSB Differential Geometry Seminar*, Santa Barbara
- Jan. 2020 **Witten deformation on noncompact manifolds**, *AMS Special Session on Differential Geometry and Global Analysis, I Honoring the Memory of Tadashi Nagano*, Denver
- Nov. 2019 **Witten deformation on noncompact manifolds**, *AMS Special Session on Topics in Global Geometric Analysis*, Los Angeles

## Teaching Activities

- Instructor** *Calculus II*, Spring 2024 at Peking University  
*Math 1342: Calculus 2 for Science & Engineering*, Fall 2024 at Northeastern University  
*Math 1342: Calculus 2 for Science & Engineering*, Winter 2025 at Northeastern University  
*Math 2331: Linear Algebra*, Fall 2025 at Northeastern University  
*Math 7371: Morse Theory*, Fall 2025 at Northeastern University
- Teaching Assistant** *Math 8: Transitions to Higher Mathematics*, Spring 2018 at UCSB  
*Math 8: Transitions to Higher Mathematics*, Fall 2018 at UCSB  
*Math 6B: Vector Calculus*, Winter 2019 at UCSB  
*Math 34B: Calculus for Social and Life Sciences*, Spring 2019 at UCSB  
*Math 6A: Vector Calculus*, Fall 2019 at UCSB  
*Math 6A: Vector Calculus*, Winter 2020 at UCSB  
*Math 6A: Vector Calculus*, Summer 2020 at UCSB  
*Math 8: Transitions to Higher Mathematics*, Fall 2020 at UCSB  
*Math 6A: Vector Calculus*, Winter 2021 at UCSB

*Math 6A: Vector Calculus, Spring 2021 at UCSB*

**Reader** *Math 117: Methods of Analysis, Fall 2017 at UCSB*

*Math 108B: Introduction to Linear Algebra, Winter 2018 at UCSB*

*Math 117: Methods of Analysis, Fall 2019 at UCSB*